

MUHAMMAD HAMZA KHAN

Software Developer | WebGIS Developer | GIS Analyst | AI-Assisted Systems
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Summary

Software developer and Geoinformatics Engineering student at NUST IGIS, working across full-stack web apps, GIS dashboards, remote sensing workflows, and AI-assisted systems. Completed an AI internship at SPS Technologies. Built 10+ projects including logistics routing, disaster intelligence, flood monitoring, solar analysis, healthcare access, and live mapping dashboards. Comfortable moving from spatial data and APIs to clear client-facing interfaces.

Experience

AI Intern | SPS Technologies 2025

Completed an AI internship focused on applying intelligent systems with a geoinformatics background, exploring practical links between AI, GIS, automation, and software workflows.

Worked with Python, automation concepts, and spatial-data thinking to understand how AI-assisted systems can support analysis, decision-making, and user-facing tools.

Project-Based Software & GIS Developer | Portfolio Builds 2024 - Present

Built 10+ apps, dashboards, GIS tools, and AI-assisted systems across logistics routing, flood monitoring, disaster intelligence, solar analysis, healthcare access, and market mapping.

Designed full-stack and map-first interfaces using React/Next.js, Python/FastAPI, PostGIS, Mapbox/MapLibre, QGIS, GDAL, REST APIs, TensorFlow, and OpenCV.

Selected Projects

RouteIQ | React | TypeScript | FastAPI | PostgreSQL | MapLibre | Python DSA 2026

Built a logistics dashboard with rider/delivery management, route simulation, and seven graph/sorting algorithms implemented from scratch, including Dijkstra, A*, TSP DP, BFS, DFS, Greedy, and Merge Sort.

HazardMind AI | Claude API | FastAPI | PostGIS | Sentinel-1/2 2026

Contributed hazard agent logic, satellite-informed risk interpretation, and geospatial reasoning to a multi-agent disaster response pipeline built with Team GridForce.

Indus Pulse Warning System | Mapbox GL JS | Node.js | WMS | REST APIs 2026

Built a flood monitoring dashboard for the Indus River basin with live map layers, alert views, operational context, and interactive WebGL visualization.

SolarVision AI | TensorFlow | Rasterio | Python | QGIS 2025

Created a satellite-based solar analysis interface with location search, configurable assumptions, and site-potential visualization for clearer evaluation decisions.

CareGrid India | FastAPI | React | RAG | PostGIS | Mapbox 2026

Supported geospatial data modeling, RAG workflow, testing, and map-backed healthcare facility intelligence; team placed 6th globally in a 21-hour hackathon build.

Remote Sensing & Cartography Lab | QGIS | GDAL | ERDAS Imagine | Spectral Indices 2024 - Present

Prepared satellite imagery, spectral indices, classification outputs, flood extent maps, vector datasets, and publication-ready map layouts.

Skills

Frontend: React, Next.js, TypeScript, JavaScript ES6+, HTML5, CSS3, Tailwind CSS

Backend: Python, FastAPI, Node.js, Express.js, REST APIs, PostgreSQL, PostGIS

GIS & Mapping: Mapbox GL JS, MapLibre, Leaflet, Deck.gl, QGIS, ArcGIS Pro, GeoServer, WMS/WFS

Remote Sensing: GDAL, Rasterio, ERDAS Imagine, GeoPandas, Shapely, SAR context, spectral indices

AI & Data: TensorFlow, OpenCV, Agentic AI, RAG pipelines, LLM integration, spatial data pipelines

Cloud & Tools: Git/GitHub, Docker, CI/CD, Vercel, Netlify, Firebase, MongoDB, AWS basics

Education

BE Geoinformatics Engineering | NUST IGIS, Islamabad

Relevant Coursework: GIS, Remote Sensing, Spatial Databases, Python, Geodetic Engineering, Statistical Analysis

Hackathons: 3 team-based builds, including CareGrid India and HazardMind AI.